

Week2 task1

Azzam fahdel alshamrani

2337310

Project Specification — Password Strength Checker

1. Objective

A command-line tool that takes a password as input and evaluates its strength (Weak / Medium / Strong), explaining which criteria it passed or failed.

2. Core Function

The application must, at minimum, successfully:

- Accept a password string from the user
- Check it against a set of basic security rules
- Output a strength rating with a clear explanation of what's missing (if any)

3. Inputs

- A password (text string) typed by the user

4. Checks to Perform (rules)

- Minimum length (e.g., 8+ characters)
- Contains at least one uppercase letter
- Contains at least one lowercase letter
- Contains at least one number
- Contains at least one special character (e.g., !@#\$%)
- (Optional, if time allows) Not in a small list of common weak passwords (e.g., "password123", "12345678")

5. Outputs

- Strength rating: Weak / Medium / Strong
- A short list of which rules passed and which failed
- (Optional) One suggestion to improve it

6. Constraints (out of scope)

- No saving/storing passwords anywhere (never store real passwords, even for testing — use throwaway examples)
- No GUI — command-line only
- No integration with real breach databases (e.g., "Have I Been Pwned" API) — that's a stretch goal, not required
- No password generation feature — checking only

7. Success Criteria

- Correctly identifies an obviously weak password (e.g., "1234") as Weak
- Correctly identifies a strong password (e.g., "Tr0ub4dor&3") as Strong
- Explanation output is clear enough that a user knows exactly what to fix